

Sean Diab

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ML Research Engineer | ML Engineer | Bay Area, CA
ML research engineer building language models and production ML systems end-to-end.

Experience & Research

Reviser – Edit-Based Language Modeling

Dec 2025 – Sep 2026

Solo Research Project | GitHub | arXiv:2609.20830 | Checkpoints | Demo

- Built 100M- and 300M-parameter edit-based language models and their PyTorch training, inference, evaluation, and trajectory-analysis systems end-to-end; models generate through autoregressive *Insert*, *Move*, and *Stop* actions over a mutable canvas.
- Ran 3,000-sample C4 pairwise LLM-judge arenas; achieved 61.3%/54.4% win rates versus matched autoregressive baselines at 100M/300M parameters and 85.9%/78.3% versus SEDD/MDLM.
- Developed obfuscation-to-restoration supervision and pooled canvas-state conditioning to produce edit-rich training data and improve state awareness without full-sequence recomputation.
- Released the code, paper, checkpoints, and interactive trajectory explorers; filed a U.S. provisional patent application covering the core architecture and training methods.

FaceOff | Founder & ML/Systems Engineer

May 2026 – Sep 2026

- Founded and solo-built native SwiftUI iOS and Jetpack Compose Android applications against a shared FastAPI and PostgreSQL backend, including live broadcasting, watching with chat, messaging and calls, Community posts, Studio media management, multi-account profiles, and replay-backed battle history.
- Built a masked-template-matching and RapidOCR pipeline over ReplayKit and MediaProjection capture that recognized matched opponents and automatically connected them through LiveKit voice rooms; reduced cross-device detection skew from 14 seconds to sub-second.
- Engineered hardware H.264 screen sharing at up to 1080p/60 fps, LiveKit/WebRTC media transport, a lower-resolution simulcast fallback, zero-re-encode recording with an ffmpeg CFR fallback, and replay clipping aligned to Supercell battle timestamps.
- Implemented Apple and Google authentication, account verification, push notifications, moderation, privacy controls, rate limiting, monitoring, backups, and release/rollback infrastructure; localized the system across nine languages and visually validated 387 locale/screen states, including Arabic right-to-left layouts.

GPT Transformer Analysis | GitHub

Mar 2025 – Jun 2025

- Authored a technical analysis of GPT forward and backward passes with complete gradient derivations and implementation checks connecting transformer theory to practical training behavior.

Publications

Reviser: Revision-Capable Text Generation via Autoregressive Cursor Actions

2026

S. Diab (sole author). *arXiv preprint* arXiv:2609.20830.

Education

University of Colorado Boulder

Jun 2024 – Feb 2025

M.S. in Computer Science; Graduate Certificate in Artificial Intelligence

GPA: 4.0/4.0

Indiana University East

Jun 2025 – Dec 2025

B.S. in Mathematics; Pi Mu Epsilon (Mathematics Honor Society)

GPA: 4.0/4.0

Skills

Languages & Tools: Python, SQL, Git, LaTeX

ML & Evaluation: PyTorch, Transformers, NumPy, language-model training and inference, LLM-judge evaluation, benchmark analysis

Backend & Infrastructure: FastAPI, PostgreSQL, asyncpg, Alembic, LiveKit, WebRTC, ffmpeg, Docker, Linux, systemd, Cloudflare Tunnel, R2

Mobile & Computer Vision: ReplayKit, MediaProjection, CallKit, APNs, FCM, OpenCV, RapidOCR

Testing: pytest, XCTest, JUnit, Maestro

Interests: Chess (approximately 2200), Clash Royale (top-100 global peak), long-distance cycling and running